

Municipal aggregates recover over four-fifths of the neighbourhood deprivation ordering that their satellite features support

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Abstract

Most countries publish social statistics as administrative aggregates; almost none publish them at neighbourhood level. We measure how much neighbourhood detail aggregates and free satellite imagery jointly recover. Trained only on five-level deprivation aggregates of Mexican municipalities — never on spatial annotation — and scored against fully held-out census-tract grades, the map reaches a within-municipality rank correlation of $\rho = 0.20$ on fourteen test cities opened once, with frozen models: 84% of what the same features attain there under full tract supervision (0.23; 90% on the validation cities used for selection). More aggregates keep buying map, state aggregates forfeit part of it, one national figure erases it, and optical content matters where resolution does not. The maps close 31–66% of the aggregate-to-census targeting gap on test, replicate against an independent institution’s index, beat the incumbent wealth index where selection occurred and match it on test, and order Bogotá’s block strata without retraining. Attention-based multiple-instance learning fails here for a measurable, structural reason.

Introduction

Census-quality social data is expensive, and most of the world sees it roughly once a decade. In between, and below the level the tables are published at, policy runs on aggregates: a municipal index here, a district average there. A rich literature has shown that satellite imagery plus machine learning predicts such indicators across space [1–4], and derived global products already steer programme targeting [3, 5]. But predicting an aggregate somewhere else and *disaggregating it in place* are different tasks, and the second carries the policy weight: a mayor knows their municipality is poor; what they lack is the map of which neighbourhoods are poorest.

We ask the disaggregation question quantitatively: how much neighbourhood detail do a country’s published aggregates, combined with free imagery, actually contain? We frame the answer as an exchange rate — so many aggregates, published at such a granularity, buy so much map — and we measure it where measurement is possible. Mexico publishes its Social Deprivation Grade (GRS) both as municipal aggregates and for 61,430 urban census tracts (AGEB) [6]; we train only on the former and score maps against the latter, held out entirely, in cities the model never saw (Fig. 1).

Three design choices keep the measurement honest. Supervision is genuinely weak: a model sees a bag of image tokens for a municipality and one vector — the population shares at each grade threshold, which is what the published aggregate encodes — and no spatial annotation of any kind. Every comparison is scored *within* municipalities, because ordering tracts across cities is mostly solved by knowing which city is poor (pooled correlations run near 0.46 against 0.23 within, at the same operating point). And an oracle trained directly on the held-out tract labels bounds what the frozen representation supports at all, so the quality of weak supervision is always reported as a fraction of a measured ceiling rather than of an imagined perfect map.

Results

The exchange rate

Figure 2 shows the measurement on a pool of 771 municipal bags — the 110 training cities plus 213 municipalities added expressly to cover the deprived half of urban Mexico that large-city samples miss (Fig. 1a; municipalities overlapping any held-out city’s ground are excluded, Methods). Every row below was scored once on the fourteen test cities with frozen, validation-selected models after all development ended; validation values are shown beside them as the selection benchmark. **Quantity** (Fig. 2a): fifty aggregates buy $\rho \approx 0.15$ within municipalities; the full national supply buys 0.196 on test against a test ceiling of 0.233 — an 84% recovery (validation: 0.227 of 0.251, 90%). The test curve replicates the rise through 400 aggregates almost exactly and flattens at the top: the last increment that validation credited to the full supply does not fully confirm. **Granularity** (Fig. 2b): relabelling the same bags with state-level aggregates cuts the map to $\rho = 0.152$ (test: 0.168 from 0.195); a single national aggregate erases it (0.009; test 0.004). What a statistics office chooses to publish bounds what its users can recover, on both splits and monotonically. **Content** (Fig. 2c): block-averaging the optical input to the radar’s 20 m effective resolution costs little on validation (0.183 $\rightarrow$ 0.172) and more on test (0.179 $\rightarrow$ 0.129), while radar sits far below both (0.006 validation, 0.084 test) — despite having been, in our feature-based phase-one baseline, the best single sensor at detecting high-deprivation tracts (per-city AUROC 0.773 against 0.747 optical). Sensor content dominates ground sampling distance, with resolution carrying a real secondary cost on test; the dissociation between predicting and localizing recurs across sensors.

Why the standard tool fails, and what fixes it

The standard weakly supervised localizer — gated-attention multiple-instance learning (AB-MIL [8], with the CLAM instance constraint [9]) — predicts municipal labels well (quadratic-weighted $\kappa = 0.62$) and produces a map at chance (AUROC 0.463; Fig. 3a; these diagnostics ran on the national-set pool). The reason is measurable: flattening the attention to near-uniform with an entropy penalty costs no bag accuracy (0.899 $\rightarrow$ 0.895), so the bag task is solvable from the plain feature mean and no gradient ever asks the attention to point anywhere.

Inverting the composition fixes it. Predicting every token and averaging the *predictions* — deep learning from label proportions [10–12] — makes the map the mechanism rather than a by-product: AUROC 0.761 under the same features, bags and labels on that national-set pool;

the swept head with 480 m of neighbourhood context reaches 0.802 on the expanded pool of Fig. 2. Because Mexican municipal grades are population-weighted aggregates of tract grades, the setting is label proportions with *known* instance weights; weighting each token by its tract’s census population is a robustness variant that performs equivalently ($\rho = 0.232$ against 0.227 on validation; 0.182 against 0.196 on test).

The sweep exposed a structural blindness worth naming. Across fourteen configurations, the rank correlation between bag-level error — the only model-selection signal weak supervision may legitimately use — and within-municipality map quality is -0.14 , while against the map’s pooled AUROC it is $+0.62$, the wrong sign for a selection criterion (Fig. 3b): honest model selection is nearly blind to the quantity that matters and mildly drawn toward worsening it. Whether this extends beyond settings where the bag label is an average of instance states is an open question, but any pipeline whose product is the map — computational pathology included — has to confront it, and it is why our benchmark ships with held-out instance labels.

External validity

Four independent checks argue the map measures deprivation rather than an artifact (Fig. 4). Against the leading downloadable product, Meta’s Relative Wealth Index [3], scored per tract on an identical universe with a city-clustered bootstrap of the paired difference (2,958 shared AGEb), the map wins in 18 of 22 municipalities: $\Delta\rho = +0.22$ [$+0.08, +0.29$]; on the test cities the difference is $+0.13$ [$-0.07, +0.26$], directionally consistent and not individually significant over 22 municipalities (Fig. 4a). A land-cover-plus-NDVI head with no learned representation reaches $\rho = 0.150$ on validation and 0.219 on test, where it is statistically indistinguishable from the headline (paired $\Delta\rho = -0.02$ [$-0.18, +0.10$], the headline winning 16 of 29 municipalities). The learned features’ advantage within municipalities is therefore split-dependent; what holds on both splits is their edge on the bag task (bag error 0.095 against 0.110) and, on test, on high-deprivation detection (AUROC 0.821 against 0.781). Much of within-municipality ordering is, evidently, land-cover-shaped — a finding that sharpens rather than weakens the exchange rate, which prices the supervision, not the encoder. Against an institution it never trained toward — CONAPO’s continuous urban marginalization index, built from a different indicator set over the same census — the map correlates at $\rho = 0.361 \pm 0.073$ within municipalities on validation and 0.257 ± 0.146 on test, in both cases at least as high as against its own five-level training construct, as a tie-free target should allow (Fig. 4b). In simulated geographic targeting with fractional funding of the marginal tract, allocating by the map instead of the municipal aggregate closes 26–43% of the gap to census-guided allocation on validation at budgets of 10% and above — losing at the tightest budget (5%), where errors at the top of the ranking are unforgiving — and 31–66% on the test cities, with no crossover there (Fig. 4c). The map is also not an administrative artifact: crossing a municipal border where deprivation does not change moves it by $+0.019$ [$-0.002, +0.155$], and tract borders inside municipalities are crossed smoothly ($+0.001$); an earlier $+0.14$ border jump traced to bag-limited neighbourhood adjacency — our inference seam, corrected, not the model’s shortcut. (The test cities contain too few same-grade cross-border pairs to repeat this control there.)

The construct travels; a different construct does not

Applied unchanged to Bogotá, the model orders Colombia’s block-level socioeconomic strata at $\rho = 0.318$ (AUROC 0.716 for the deprived strata 1–2 over 14,132 tokens joined to blocks; Fig. 4d). Applied to Rio de Janeiro, it detects IBGE’s mapped informal settlements barely above chance (AUROC 0.557 among built-up tokens; Fig. 4d). The contrast is informative rather than disappointing: Colombian strata measure the same construct as the GRS — a graded deprivation ordering over the whole urban fabric — while the AGSN mask delimits informality, a partly orthogonal notion. Weak supervision on aggregates taught a deprivation gradient, and the gradient, not a country signature, is what transfers.

Limits

The ceiling is the honest one: even with full tract labels and neighbourhood context, 160 m tokens support $\rho = 0.251$ within municipalities — useful for ordering neighbourhoods, far from reading individual blocks. Agreement depends strongly on tract size: $\rho = 0.26$ for tracts covered by 1–3 tokens, rising to 0.43 at 20–49 tokens, and collapsing to 0.13 in the two municipalities dominated by very large tracts; the test split reproduces the pattern noisily, peaking at 0.38 and collapsing in its single large-tract municipality — the modifiable areal unit problem stated in units an office can act on. Ecological caution applies throughout: the model recovers where deprived tracts are, and tract grades are themselves aggregates over households; neither the labels nor the maps identify deprived households, and the targeting simulation counts people in deprived tracts, not deprived people.

Discussion

The exchange rate reframes a familiar technology. Rather than another demonstration that imagery predicts poverty, it answers the question a statistics office would ask: *what does the resolution of publication buy at the resolution of inference?* The measured axes give the answer operational shape — the national supply of municipal aggregates recovers 90% of what these features support, coarser publication forfeits map in direct proportion, optical content rather than resolution is the binding sensor property — and the targeting experiment prices the result in people reached, crossover included. The negative results carry equal weight: attention-based MIL fails here for a structural reason; aggregate-level validation is nearly blind to map quality; and informality detection is not a free by-product of a deprivation gradient. Each is a warning any weakly supervised mapping effort will meet.

Mexico’s dual-level publication made the calibration possible; the measured fractions are what travel to countries that publish only aggregates. Bogotá is the first step of that argument. Extending the dual calibration to Colombia’s own strata — fine truth held out, coarse aggregates built from it — is the natural next measurement, and the released benchmark makes every number here re-runnable from frozen features in minutes.

Methods

Imagery. Sentinel-2 L2A and Sentinel-1 RTC annual median composites for 2020 (10 m grid, windowed COG reads from Microsoft Planetary Computer), for 424 boxes covering the 138 cities of the national set and the 213-municipality expansion; city boxes are conurbations and typically span several municipalities. **Labels.** CONEVAL GRS 2020: municipal-level population shares at each of four grade thresholds supervise training; AGEB-level grades are used only for evaluation and for the oracle. Auxiliary evaluation sources: CONAPO IMU 2020 [7], ESA WorldCover 2020 [14], Meta RWI [3], IBGE AGSN 2019 [15], Bogotá’s district stratification [16]. **Bags and instances.** A bag is a municipality (771 after exclusions); instances are 16-pixel tokens (160 m) encoded by a frozen DOFA foundation model [13], both sensors fused per token. Municipalities contributing ground to any held-out city are excluded from the expansion — an earlier version leaked nine validation municipalities this way, detectable only at bag level, and every affected number was re-measured. **Model.** A two-layer head on standardized frozen features predicts four cumulative-grade probabilities per token; the bag prediction is the mean of token predictions, fit with binary cross-entropy; each token’s hidden representation is concatenated with the mean of its 8 grid neighbours (480 m), computed city-wide at inference. **Protocol.** 110/14/14 cities for train/validation/test, stratified by size and deprivation, no city split across sets. The test cities were opened exactly once, after all development ended: every configuration retrains with early stopping anchored to the validation cities as during development (deterministic given the seed), or loads its saved weights, and the test bags are only ever scored under **inference mode**. Nothing was tuned, selected or iterated against test results. Sweeps select on grouped training folds under bag error only; validation cities are scored once per reported configuration; comparisons are paired by municipality over three seeds; every interval on a within-municipality mean is a city-clustered bootstrap (2,000 resamples), and curve error bars are s.d. over seeds. **Oracle.** The same head trained on tract labels, with the same context, bounds the representation ($\rho = 0.251$ with context, 0.202 without). **Zero-shot transfer.** Bogotá and Rio get the same 2020 composite protocol; tokens are scored by the three final seeds with Mexican standardization statistics, deliberately unadapted. Rio’s evaluation is restricted to tokens with WorldCover built fraction ≥ 0.10 . **Sweep.** Fourteen configurations varied one ingredient each (learning rate 5e-5–1e-3, hidden width 64–512, dropout 0–0.5, weight decay, loss, input standardization), then a combination round; standardized inputs with binary cross-entropy won on bag error over five grouped training folds. **Targeting simulation.** For each validation city and budget $q \in \{5, 10, 20, 30\}\%$ of population, tracts are funded in the order given by each allocator — municipal aggregate (ties broken at random, 200 draws), the map’s tract-mean score, or tract truth — with the marginal tract funded fractionally, so the truth ordering upper-bounds both by construction. The outcome is the population living in high or very-high grade tracts among those reached, pooled over cities before the gap ratio is formed. **RWI pairing.** Each AGEB takes the RWI value of the nearest grid point to its centroid; the model is averaged token-to-tract; both are scored on the 2,958 shared AGEB in the 22 municipalities with at least 20 scored tracts and more than one grade, and the interval is a city-clustered bootstrap of the per-municipality difference. **MAUP bins.** Tracts are binned by tokens contained (1–3, 4–9, 10–19, 20–49, ≥ 50); within-municipality correlations use municipalities with at least ten tracts in the bin. **Border**

test. Adjacent token pairs (160 m apart) are classed by what their midpoint crosses; the excess mean absolute score jump of same-grade municipal-border pairs over same-tract pairs is city-bootstrapped; same-grade AGEB-border pairs inside one municipality are the placebo. **Data and code.** The full pipeline, frozen vectors, labels, splits and protocol are released; every figure regenerates from one canonical results file.

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Data availability

The frozen benchmark — foundation-model vectors for every bag, municipal and tract labels, splits and the evaluation protocol — will be deposited on Zenodo with a DOI at publication. All source data are public: Copernicus Sentinel via Microsoft Planetary Computer, CONEVAL, CONAPO, INEGI, ESA WorldCover, Meta RWI (HDX), IBGE, and Bogotá’s open-data portal.

Code availability

The full pipeline (MIT licence) is available at the project repository, pinned by `uv.lock`; each figure maps to one driver script.

Author contributions

D.P.A. conceived the study, built the pipeline, ran the experiments and wrote the manuscript.

Competing interests

The author declares no competing interests.

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Figures

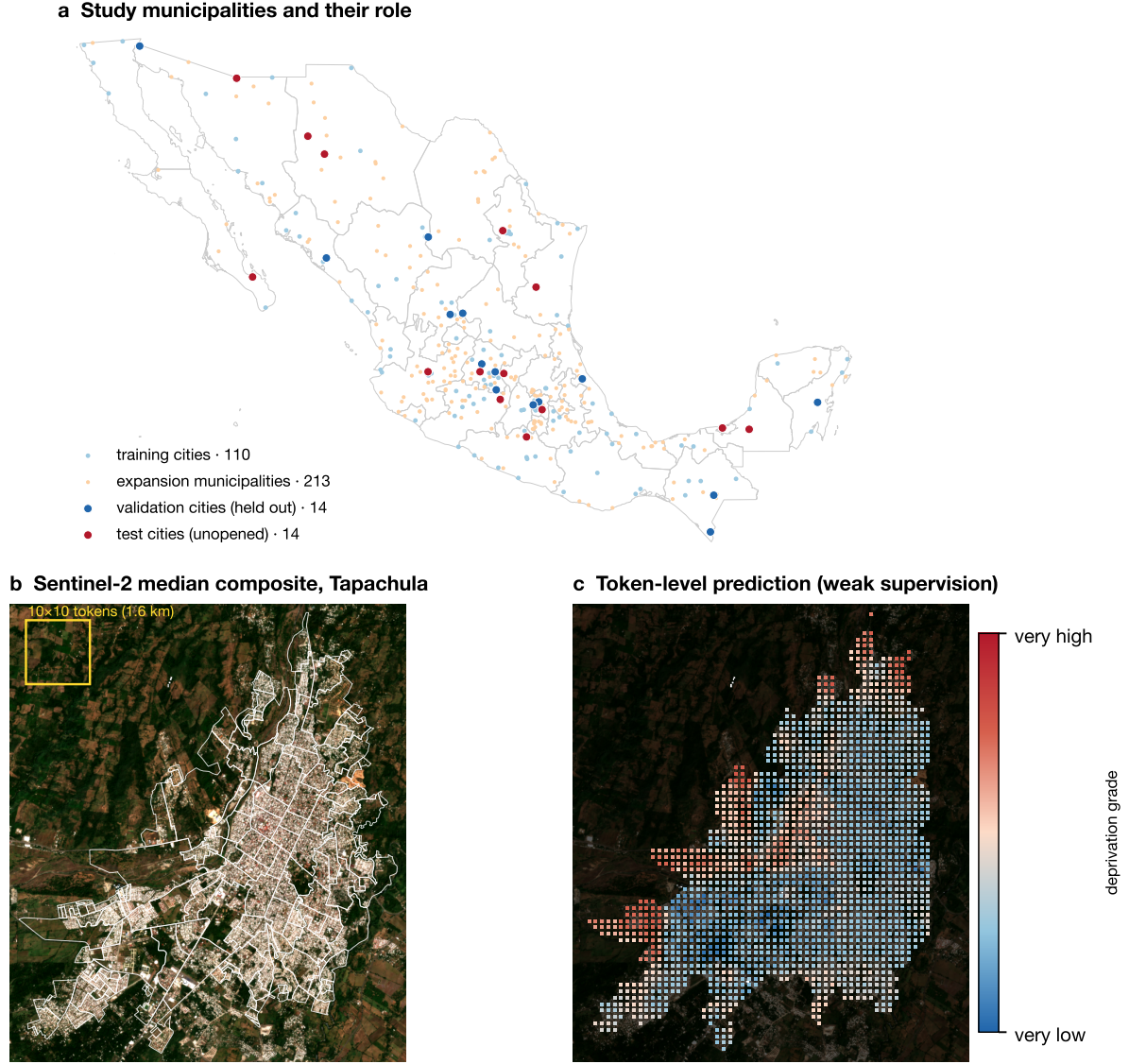


Figure 1: **Study design.** **a**, The 351 study units: 110 training cities, the 213-municipality expansion toward the deprived half of urban Mexico (municipalities overlapping held-out ground excluded), 14 validation and 14 unopened test cities. A city box is a conurbation and typically yields several municipal bags, which is how 351 units produce 771 bags. **b**, What the model consumes: the 2020 Sentinel-2 median composite of Tapachula (a validation city) with AGEB boundaries overlaid; tokens of 160 m tile the box. **c**, What it produces: the token-level deprivation score from municipal aggregates alone. Tract ground truth for the same city is shown in Supplementary Fig. 1; Tapachula sits at the validation median ($\rho = 0.47$), chosen so the example neither flatters nor sandbags the method.

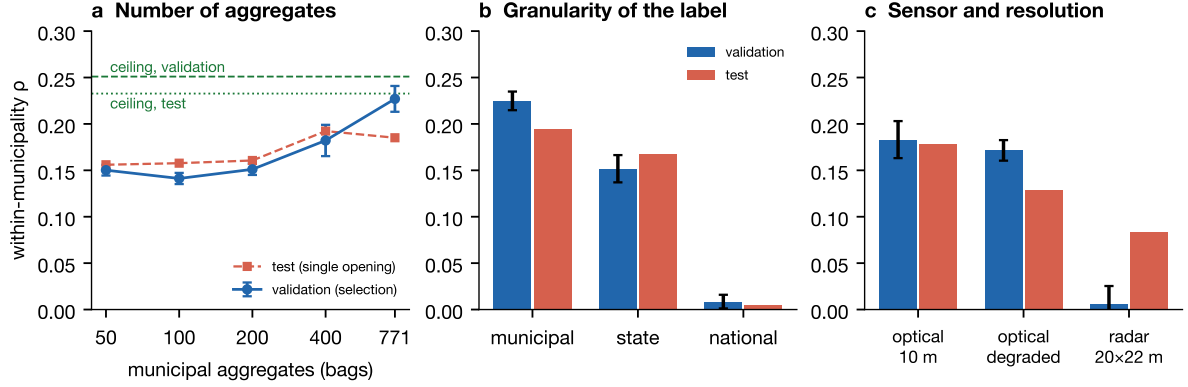


Figure 2: **The exchange rate between aggregates and maps.** All panels: mean over three seeds against held-out tract grades; blue, the 14 validation cities used for selection (error bars, s.d. over seeds); orange, the 14 test cities scored once with frozen models; dashed and dotted lines, the instance-supervised oracle on each split ($\rho = 0.251$ validation, 0.233 test). **a**, Agreement as a function of how many municipal aggregates supervise training (nested, grade-stratified sampling). **b**, The same 771 bags relabelled with coarser aggregates. **c**, Single-sensor configurations: destroying optical resolution to the radar’s grain costs little on validation and a real but secondary amount on test; radar sits far below both.

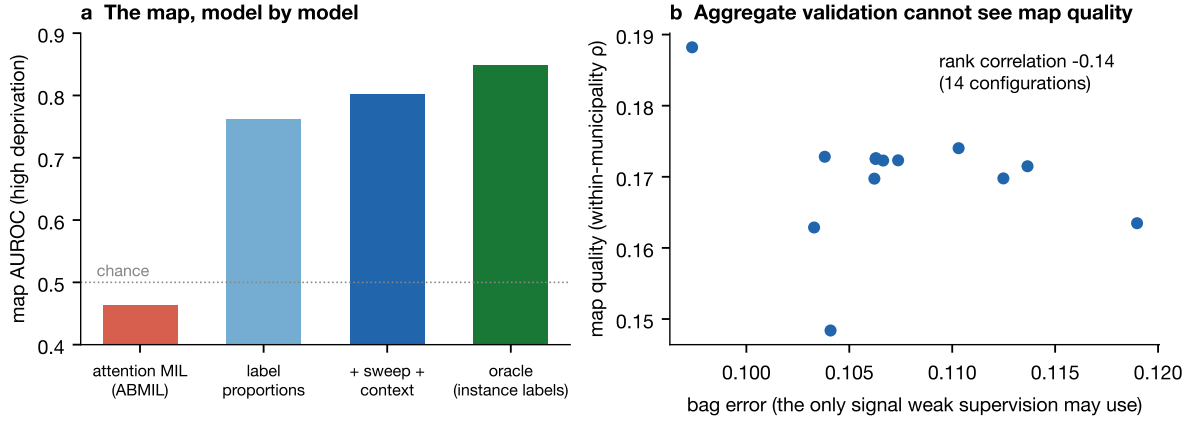


Figure 3: **Predicting is not localizing.** **a**, Map quality by model family on identical features and labels; the first three bars are measured on the national-set pool, the swept bar on the expanded pool of Fig. 2. Attention pooling solves the bag task from the feature mean and receives no localization gradient; predicting instances and averaging predictions makes the map the mechanism. **b**, Across the fourteen sweep configurations (grouped training folds of the national set), the only criterion weak supervision may monitor is nearly uncorrelated with within-municipality map quality — and positively correlated (+0.62) with the pooled AUROC it would seem to proxy.

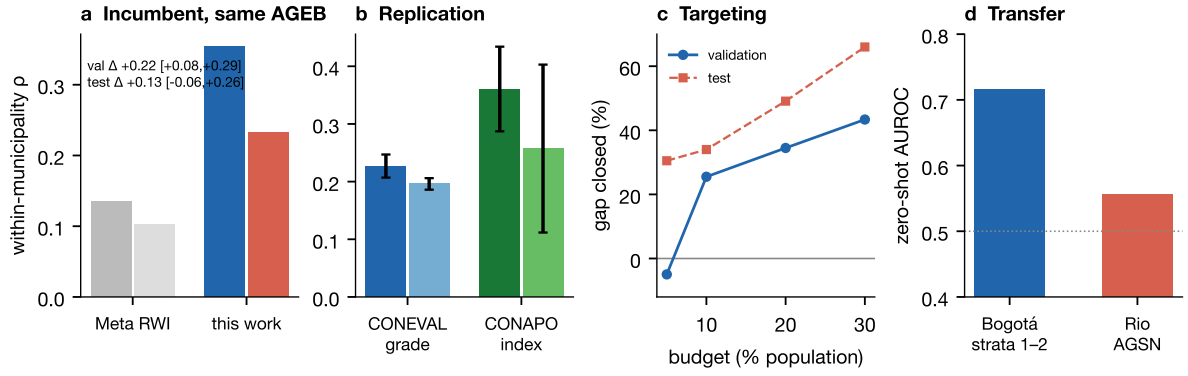


Figure 4: **External validity.** Blue/dark bars, validation; orange/light bars, test (single opening). **a**, Within-municipality agreement on the identical AGEB universe reached by the incumbent’s grid; annotation, city-clustered bootstraps of the paired difference per split. **b**, Replication against an independent institution’s continuous index (error bars, city-clustered bootstrap half-widths). **c**, Share of the aggregate-to-census targeting gap closed at each budget, people pooled per split; validation loses at the tightest budget, test does not. **d**, Zero-shot transfer: the Mexican model, unchanged and unadapted, ordering Bogotá’s block strata (AUROC for strata 1–2; $n = 14,132$ tokens) and detecting Rio’s mapped informal settlements among built-up tokens ($n = 34,331$). The deprivation gradient travels; informality is a different construct.